

Total No. of Questions : 4]

SEAT No. :

PA-10229

[6010]- 105

[Total No. of Pages : 2

**B.E. (Mechanical Engineering) (Insem)**  
**RENEWABLE ENERGY TECHNOLOGIES**  
**(2019 Pattern) (Semester - VIII) (Elective - VI) (402051 B)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

- Q1)** a) Write short note on: JNNSM policies and initiatives. [5]  
b) Define solar irradiance, solar constant, extra-terrestrial and terrestrial radiance? What is the standard value of solar constant? [5]  
c) Calculate the angle made by beam radiation with the normal to a flat plate collector on 31<sup>st</sup> March at 10.00 am solar time. The collector is located at Bombay (19° 07' N and 72° 51' E) is tilted at an angle of 30° with the horizontal and pointing due south. [5]

OR

- Q2)** a) Bring out the significance of utilization of renewable energy sources in the context of world energy scenario. [5]  
b) With the help of a neat sketch explain the solar pyrheliometer. [5]  
c) A propeller type wind turbine has following data [5]  
Speed of wind at a height of the 10 m = 12 m/s  
Air density = 1.226 kg/m<sup>3</sup>  
Height of tower = 100 m  
Diameter of rotor = 80 m  
Power coefficient = 0.35  
Transmission efficiency = 80%  
Generator efficiency = 85%  
Find:  
i) Total power available in wind  
ii) Power extracted by wind turbine  
iii) Electrical power generated

P.T.O.

- Q3) a)** Calculate heat loss from the flat plate collector with two glass covers with following data: [7]

Size of absorber plate = 1.9 m × 0.9 m

Height of collector casing = 16 cm

Mean plate temperature = 70° C

Ambient air temperature = 24° C

Back insulation thickness = 8 cm

Side insulation thickness = 4 cm

Thermal conductivity of insulation = 0.05W/mK

Convective heat transfer coefficient between top cover and the surrounding air = 5.7 W/m<sup>2</sup> K

Emissivity of glass cover = 0.88

Assume temperature of sky ( $T_{sky}$ ) is 6°C lower than ambient temperature and temperature of top glass cover as 32°C.

- b) Classify concentrating type of collector. Discuss their advantages and disadvantages. Explain construction and working of any one type of concentrating collector. [8]

OR

- Q4) a)** Draw a neat sketch of solar air heater and explain its principle of working. [5]
- b) Solar drying is superior to traditional drying, explain in brief. What are different types of solar driers? Enlist any three applications. [5]
- c) Explain with neat sketch procedure for testing liquid flat plate collector. [5]

